

SARA ARIF

+92-336-3124697 | saraarif6198@gmail.com | [linkedin.com/in/sara-arif-792p](https://www.linkedin.com/in/sara-arif-792p) | github.com/SaraArif6198 | sara-3d-portfolio.vercel.app

PROFESSIONAL SUMMARY

Production-oriented Computer Science undergraduate (CGPA 3.85/4.0, ranked 2nd in department) with shipped systems spanning full-stack development, ML pipelines, data engineering, and multi-agent AI. Delivered 10+ live projects — from an offline-first PWA serving 50+ field workers to LALR(1) compiler simulators and LangGraph agent backends. Silver Honour — Top 2% globally, ICSC 2025 (3,800+ participants).

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL (MySQL, PostgreSQL), C++, C#, PHP

AI / ML: Scikit-learn, CatBoost, SHAP, SMOTE, TensorFlow, LangGraph, RAG, HuggingFace, Groq, Streamlit

Data & Analytics: Pandas, NumPy, ETL Pipelines, Power BI, Advanced DAX, Tableau, Snowflake, Microsoft Fabric

Web & Mobile: React, React Native, FastAPI, REST APIs, Progressive Web Apps, PHP/MySQL, Service Workers

Tools: Git, Docker, Vercel, SQLite, Figma, Cisco Packet Tracer

TECHNICAL PROJECTS

VizBot Analytics (DataSage) · *LangGraph · Groq Llama 3.1 · FastAPI · Python*

- Orchestrated a multi-agent LangGraph pipeline processing natural-language EDA queries — data profiling → anomaly detection → visualization → narrative — reducing manual analyst workflow to zero user-side steps.
- Decoupled LLM inference into an async FastAPI backend, eliminating UI-blocking latency and enabling concurrent multi-user query execution with sandboxed code isolation.

Bank Loan Risk Classifier · *Python · CatBoost · SHAP · SMOTE · Streamlit*

- Resolved a 90/10 class imbalance via SMOTE; tuned CatBoost classifier achieving 30%+ reduction in false negatives on loan default predictions — directly lowering business-critical credit risk.
- Deployed interactive Streamlit dashboard with per-prediction SHAP waterfall plots, cutting non-technical stakeholder model review time by 50%.

Customer Trends ETL Pipeline · *Python · SQL · Pandas · Power BI · Advanced DAX*

- Engineered full ETL pipeline — SQL extraction → Python feature engineering → Power BI executive dashboard — automating a previously manual weekly reporting workflow and compressing turnaround from days to real-time refresh.

UrduPy Compiler · *Python · Tkinter · Compiler Theory*

- Designed a complete 5-phase compiler (Lexer → Parser → AST → IR → CodeGen) for Roman Urdu from scratch, targeting 230M+ Urdu speakers with zero prior programming tooling in the language.
- Packaged with a custom Tkinter IDE featuring real-time type inference, syntax highlighting, and inline error reporting — delivering a complete developer experience end-to-end.

SQL2Pandas Transpiler · *Python · AST Parsing · Compiler Optimization*

- Built a 6-phase SQL-to-Pandas transpiler supporting 20+ SQL constructs with 5 optimizer passes (predicate pushdown, projection pruning, constant folding), producing efficient, readable output code.

WORK EXPERIENCE

Freelance Software Architect — Self-Employed (Remote)

May 2026 – Present

- Architected and shipped the DG Cement Work Tracker — a production PWA replacing 100% of paper-based payroll and attendance tracking for 50+ construction workers, with 28 REST API endpoints and offline-first service worker caching for remote-site operation.
- Automated end-to-end payroll processing workflows, eliminating manual data entry errors and reducing administrative processing time by an estimated 70%+ per weekly cycle.

Machine Learning Intern — DevelopersHub Corporation (Remote, Full-Time)

Jun 2025 – Jul 2025

- Conducted production EDA on raw business datasets; applied SMOTE + CatBoost hyperparameter tuning to achieve 30%+ false-negative reduction on loan default classification.
- Implemented SHAP explainability layer generating stakeholder-ready interpretability reports, reducing model review cycles by 50% and enabling non-technical audit of predictions.

EDUCATION

University of Chakwal — B.S. Computer Science

Sep 2023 – Sep 2027 (Expected)

CGPA 3.85/4.0 | Ranked 2nd in CS Department | PM Laptop Scheme 2025 — Ranked 27th / 6,000+ students

CERTIFICATIONS

Generative AI Professional (Snowflake) | Advanced Deep Learning Specialist (IBM/Coursera) | AI Agents: RAG & LangChain (IBM/Coursera)
ChatGPT Advanced Data Analysis (Vanderbilt/Coursera) | Microsoft Fabric (Whizlabs/Coursera) | Building Voice Agents with Vapi (DataCrumbs)

ACHIEVEMENTS

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- Silver Honour — Top 2% globally at ICSC 2025 among 3,800+ international participants.
 - PM Laptop Scheme 2025 — Ranked 2nd in CS Department, 27th university-wide out of 6,000+ students; merit-awarded by the Government of Pakistan.